

Redundancy across Connectives*

Alexandros Kalomoiros
Heinrich Heine Universität Düsseldorf

Matthew Mandelkern
New York University

Paul Marty
Universidade de Lisboa

Jacopo Romoli
Heinrich Heine Universität Düsseldorf

Florian Schwarz
University of Pennsylvania

Abstract. We present three experiments investigating redundancy across connectives. Specifically, we examine: (i) *Hurford disjunctions*—disjunctions where one disjunct entails the other (Hurford 1974); (ii) *Hurford conditionals*—conditionals corresponding to Hurford disjunctions (Mandelkern & Romoli 2018); and (iii) the corresponding conjunctions in which one conjunct entails the other. Our study addresses three questions about redundancy. First, does order matter? For example, are $p^+ \vee p$ and $p \vee p^+$ judged as equally infelicitous? Second, does polarity matter? For example, are $p^+ \vee p$ and $\neg p \vee \neg p^+$ judged equally unacceptable? Third, does the choice of connective affect the strength of oddness judgments arising from redundancy? Finally, we consider the broader implications of our findings and how they might be explained within existing theoretical frameworks.

Keywords: Hurford disjunction, Hurford conditional, local contexts, triviality, simplification, redundancy, super-redundancy, communicative principles

Word Count: 13761 words

* Acknowledgments: We would like to thank Patrick Elliott, Raven Zhang, the members of the Penn Meaning Lab and of the Düsseldorf semantics colloquium, as well as audiences at ZAS, IJN and SuB29 for helpful feedback.

1 Introduction

Informativity is a core communicative principle: when asserting a sentence, a speaker must convey information that is not already part of the interlocutors' common ground (Stalnaker 1974). This principle is illustrated in the contrast between (1) and (2).

- (1) CONTEXT: *we know that John lives in France.*
#John lives in France.
- (2) CONTEXT: *we have no idea where John lives.*
John lives in France.

In a standard approach modeling the context as a set of possible worlds, and assertion as a proposal to intersect the context C with the proposition expressed by a sentence S (Stalnaker 1978), the informativity constraint can be expressed as follows:

- (3) **Global informativity:**
A sentence S is infelicitous in a context C if $C \cap \llbracket S \rrbracket = C$

The constraint in (3) is global because it refers to S as a whole, without any consideration of the role of potential parts of complex sentences. As is well known, however, global informativity alone is insufficient to account for seemingly related intuitions about oddness. Consider the contrast below:

- (4) CONTEXT: *we have no idea where John lives.*
 - a. John lives in Paris.
 - b. #John lives in Paris and he lives in France.

Taken as a whole, (4a) and (4b) are contextually equivalent, as their assertion leads to the intersection of the proposition that John lives in Paris with the context, so (3) treats them on par. Intuitively, however, (4b) is infelicitous, because the second conjunct is *locally* uninformative, since residing in Paris entails living in France. Local violations of informativity also appear in the disjunctive counterpart of (4b) below, an instance of so-called 'Hurford Disjunctions' (Hurford 1974; see Marty & Romoli 2022 for an overview):

- (5) CONTEXT: *we have no idea where John lives.*
 - a. John lives in France.
 - b. #Either John lives in France or he lives in Paris.

Paralleling the above, the disjunction in (5b) conveys the same information as (5a) in context (i.e., the proposition that John lives in France), so it is not ruled out by

(3). However, the second disjunct seems locally uninformative as it does not add anything to the meaning of the overall disjunction.

To account for such local violations of informativity, two major theoretical approaches have been proposed in the literature: the Simplification approach (Katzir & Singh 2013, a.o.) and the Local Context approach (Schlenker 2009, a.o.).

The basic intuition behind Simplification is that a sentence is odd when it is equivalent to one of its own parts. More precisely, on this approach, a sentence S is deemed odd whenever it contains a constituent D that is redundant—in the sense that D could be omitted without affecting meaning, i.e., if S' is obtained from S by deleting D , and S and S' are contextually equivalent. For example, a conjunction of the form $p^+ \wedge p$ (where p^+ entails p) is odd because it is contextually equivalent to its stronger conjunct, p^+ . Similarly, a disjunction of the form $p \vee p^+$ is odd because it is contextually equivalent to its weaker disjunct, p .

The intuition behind the Local Context approach is that a sentence is odd when it has a part which is either entailed by or inconsistent with its *local context*. There are different notions of local context in the literature, but all agree that the local context for q in $p \wedge q$, in context C , is $C \cap p$, while the local context for q in $p \vee q$ in C is $C \cap \neg p$. Hence, on this approach, $p^+ \wedge p$ is odd because p is entailed by its local context, while $p \vee p^+$ is odd because p^+ is inconsistent with its local context (which generally is also taken to be a violation of constraints on informative assertions).

In this paper, we experimentally test three questions in relation to these theories. These questions stem from, sometimes controversial, intuitive judgments reported in the literature; thus, shedding experimental light onto these controversial judgments is of some importance and the results may end up requiring theoretical refinements or revisions to currently established perspectives. The first question concerns order. While there is widespread agreement in the literature that $p^+ \wedge p$ is infelicitous, there is disagreement about the status of sentences with the reverse order, $p \wedge p^+$; some authors consider these (more) acceptable, whereas others take them to be equal to $p^+ \wedge p$ in terms of their felicity, (see Katzir & Singh (2013)). By contrast, there seems to be consensus that both orders of Hurford disjunctions are infelicitous. In light of the potential role of order for at least some connectives, we thus ask:

Q1 EFFECT OF ORDER: Can the order of clauses in complex sentences impact judgments about oddness when two sentences are otherwise equivalent?

The basic setup of the Simplification and Local Context (LC) approaches is compatible with either outcome for this question: they both can be fleshed out in asymmetric and symmetric variants, which respectively do or do not yield a role

for order. While a quite natural asymmetric version of the LC approach only takes into account preceding material, a symmetric version, which takes into account both preceding and following material in a complex sentence, can also be formulated. The version of Simplification informally presented above is symmetric, but to obtain an asymmetric variant, one can replace the notion of contextual equivalence above with a notion of contextual equivalence holding fixed *just what is to the left of the deleted constituent* (Mayr & Romoli 2016). However, since the choice points of asymmetry vs. symmetry are made at a global level, independent of any particular expressions, an important feature across these variants is that order effects are predicted to be *uniform across different connectives*: either there are no asymmetries across connectives, or all connectives display asymmetries.

Another dimension of complexity and controversial intuitions reported in the literature concerns the issue of whether the presence of negation impacts relevant judgments of oddness, even when the relevant contextual entailment relations remain fixed. For example, Kalomoiros (2023a) claims that (6b) is less infelicitous than (6a), even though in both cases, the second disjunct entails the first.

- (6) a. # Either John lives in France or he lives in Paris.
- b. #? Either John doesn't live in Paris or he doesn't live in France.

Similar issues arise for another case of related oddness-effects due to redundancy, namely so-called *Hurford conditionals* (Mandelkern & Romoli 2018). These can be constructed from Hurford disjunctions by applying what we'll refer to as the *or-to-if* inference pattern, whereby (7b) follows from (7a):

- (7) a. #Either John lives in Paris or he lives in France.
- b. #If John doesn't live in Paris, he lives in France.

Parallel to the above claim for disjunction, Mandelkern & Romoli (2018) claim that (8b), the conditional derived from the negated Hurford disjunction, is less infelicitous than (7b) (and perhaps fully felicitous):

- (8) a. #? Either John doesn't live in France or he doesn't live in Paris.
- b. If John lives in France, he doesn't live in Paris.

This is remarkable insofar as, at some level of abstraction, they both have the form 'If p , then $\neg p^+$ '. (To see that (7b) has this form logically speaking, note that it is equivalent to 'If John lives outside of Paris, then he doesn't live outside of France'.)

Neither the Simplification nor the LC approach predicts a role for the presence or absence of negation per se, since on both perspectives, all that should matter is (contextual) entailment relations. So, if these judgments are empirically confirmed, the theories are missing important generalizations about the role of negation in

oddness judgments, as things stand. Hence our second goal is to explore these contrasts and see whether the presence or absence of negation (holding fixed contextual entailment relations) impacts oddness judgments in disjunctions and conditionals:

Q2 EFFECT OF NEGATION: When contextual entailment relations are held fixed, does negation matter for the purposes of computing redundancy?

Given that our experimental data will offer a first quantitative perspective on redundancy-based oddness judgments, we will also ask whether there are differences in the effects across connectives. Such differences could show up as general differences in strength, or as differences in the potential role of the other factors modulating redundancy effects:

Q3 EFFECTS OF CONNECTIVE:

- a. Does the strength of redundancy-based oddness judgments vary across connectives?
- b. Does the role of the factors that (potentially) modulate the strength of redundancy-based oddness judgments (i.e., order and polarity) vary across connectives?

Our main findings are as follows:

- We confirm the standard intuitions reported in the literature: sentences of the form $p^+ \wedge p$, $p^+ \vee p$, $p \vee p^+$, and ‘if $\neg p^+$, p ’ are degraded.
- We uncover a striking asymmetry in the role of order across connectives. While Hurford disjunctions are equally degraded in both orders, conjunctions display an asymmetry: $p^+ \wedge p$ is degraded, but $p \wedge p^+$ is not. Conditionals exhibit the most complex pattern, with surprising interactions indicating that the impact of order on redundancy is further modulated by polarity.
- Redundancy judgments in disjunction and conditionals are polarity-sensitive, in line with earlier intuition-based claims in the literature.
- The strength of redundancy effects differs across connectives.

The rest of this paper is structured as follows: Sections 2 and 3 present our first two experiments, which investigate disjunction and conditionals, respectively, manipulating both polarity and order; Section 4 then directly compares these two connectives. Section 5 reports an experiment on conjunction, focusing on order, and Section 6 situates conjunction within a broader comparison to disjunction

and conditionals. Finally, Section 7 discusses the theoretical implications of our findings.

2 Experiment 1: Disjunction

Our first experiment uses an acceptability judgment task to explore effects of redundancy, and potential modulations of such effects by negation and order.

2.1 Participants

We recruited 199 participants from a subject pool at the university of one of the authors; participants received class credit for their participation. All participants were undergraduate students who were native English speakers.

2.2 Design

The critical conditions in our design consisted of Hurford Disjunctions, where one disjunct asymmetrically entails the other. We included both disjunctions with positive disjuncts and negated disjuncts, and also manipulated their order. HD (Positive Hurford Disjunction) refers to standard cases without negation and NEGHD (Negative Hurford Disjunction) refers to cases where the disjunction contains negation. When the first disjunct is stronger, and thus would be redundant if it were (symmetrically) considered in light of the second disjunct, the disjunction is labeled FIRST; when it comes second, it is labeled SECOND.

Each item consisted of a sentence presented with a preceding context that motivated the use of a disjunction overall, and that ensured the disjuncts were plausible, salient alternatives. Participants' task was to indicate how natural the sentence was in the given context (see below for more details on the procedure and trial structure). The contexts for disjunctions containing negation, in particular, were designed to enhance the naturalness of using negated disjuncts, which may otherwise be marked and less acceptable.

(9) *Positive Hurford Disjunction* (HD)

CONTEXT: John and his family want to visit Asia. They have various destinations in mind, but the most prominent ones are cities in Japan, especially Tokyo. Recently, John and his family returned from their Asia trip, so I thought:

- a. Either John's family visited Tokyo or they visited Japan. (FIRST)
- b. Either John's family visited Japan or they visited Tokyo. (SECOND)

(10) *Negative Hurford Disjunction* (NEGHD)

CONTEXT: John and his family want to visit Asia. They have various destinations in mind like some cities in Japan (especially Tokyo), as well as some cities in China. However, I have no idea if they could afford to go to all of these places. Recently, John and his family returned from their Asia trip. I noticed that they made no mention of Tokyo or Japan, so I thought:

- a. Either John's family didn't visit Japan or they didn't visit Tokyo.
(FIRST)
- b. Either John's family didn't visit Tokyo or they didn't visit Japan.
(SECOND)

To investigate the questions above, we aimed, first, to experimentally measure the impact of redundancy, and furthermore to assess potential modulating roles of negation and order. Concretely, we tested (i) whether the presence of an entailment relation between the two disjuncts decreases the naturalness ratings of these sentences in context; (ii) whether the presence of negation modulates this effect; and (iii) whether the order of the disjuncts influences any observed effects. Our analyses also take into account the possibility that these factors might interact.

So far, our design manipulates two factors, namely ORDER and POLARITY. We were also interested in assessing the effects of these factors on redundancy. This requires appropriate baseline conditions for comparison, which introduce no grounds for potential redundancy violations but are otherwise maximally similar to the critical sentence variants. Thus, we included the following Simple Disjunction (SD) control conditions in variations parallel to those above; this 'Simple' vs. 'Hurford' manipulation constituted our third experimental factor, REDUNDANCY.

(11) *Positive Simple Disjunction* (SD)

CONTEXT: John and his family have been wanting to visit Asia for a long time. They considered various options, but in the end had to choose between going to Tokyo in the spring, or visiting some place in China for vacation in the summer. I have no idea what they ended up choosing, but given that the summer has ended now, I know that:

- a. Either John's family visited Tokyo or they visited China. (FIRST)
- b. Either John's family visited China or they visited Tokyo. (SECOND)

(12) *Negative Simple Disjunction* (NEGSD)

CONTEXT: John and his family have been wanting to visit Asia for a long time. They considered various options and initially thought about going both to Tokyo and to some places in China this summer. I have no idea if they had time to do either the Tokyo or the China option. However, I'm

sure that they couldn't afford to go to both. So, even though I don't know where they traveled, I know that:

- a. Either John's family didn't visit China or they didn't visit Tokyo.
(FIRST)
- b. Either John's family didn't visit Tokyo or they didn't visit China.
(SECOND)

These baseline conditions eliminate the entailment relationship between the disjuncts (here, for instances, by replacing 'Japan' with 'China'). This allows us both to experimentally measure the impact of redundancy and control for any potential order- or negation-based effects on the acceptability of disjunctions that arise independently of entailment. For instance, suppose that we observe a decrease in the acceptability of HD-FIRST compared to HD-SECOND. If this decrease exceeds the order effects observed in SD-FIRST versus SD-SECOND, we can conclude that the effect in the HD conditions is specifically linked to redundancy, rather than general order effects found in standard disjunctions. The NEGSD controls play a similar role. First, they assess independent order effects in negated disjunctions, ensuring that any observed effects are not solely due to word order. Second, they evaluate the impact of negation, independent of whether an entailment relationship exists between the disjuncts.

The stimuli outlined above represent our experimental conditions. In total, 24 items were created.¹ Additionally, we included 24 fillers, which are omitted here for brevity (see the supplemental materials on the OSF page for this paper, linked [here](#), for a full list of stimuli).

2.3 Procedure

The REDUNDANCY factor was implemented as a between-subjects manipulation, and the ORDER and POLARITY factors were manipulated within-subjects. As a result, the items were divided into two main lists. List 1 included the four baseline conditions corresponding to the SD and NEGSD items presented in both orders. List 2 included the four target conditions corresponding to the HD and NEGHD items in both orders. For each list, there were counterbalanced variants that included 6 critical items per condition (representing the 4 combinations of the 2 factor levels of ORDER and POLARITY), yielding 24 critical items, and 24 fillers, hence 48 items in total.

¹ 'Item' here denotes a collection of contexts and test sentences that form a coherent whole. For example, the sentences above involving John and his family visiting (or not visiting) Japan are instantiations of one item, which might be thought of as the *John's family (not) visiting Japan* story/item. In total 24 such stories/items were created and were used to create instantiations of each condition within that story/item.

After providing informed consent, participants completed three training trials to familiarize themselves with the task. Each training trial presented a context followed by a sentence. The sentences included: (1) a simple conjunction with no redundancy violation and no negation, (2) a conjunction where the first conjunct entailed the second (i.e., a redundancy violation), and (3) a simple conjunction with no redundancy violation but with negated conjuncts. Participants rated the naturalness of each sentence in context on a 7-point scale (1 = least natural, 7 = most natural) and received feedback on their responses. The experiment then began, with items presented in a randomized order within each group. In each trial, the context was presented first, and participants had to press the space bar to proceed to the sentence, which was only possible after two seconds. This was intended to ensure that they would read the context before turning to the sentence.

2.4 Data availability

All stimuli, together with the files and scripts required for the statistical analysis reported below, can be found on the *OSF* page for this paper, linked [here](#).

2.5 Data analysis

Our main objective was to measure effects of redundancy, that is, to test whether Hurford Disjunctions (HDs) are less felicitous than their Simple Disjunctions (SD) baseline counterparts. We also wanted to test whether polarity and order modulate redundancy effects. To allow for potential interactions of these factors, we thus started from a 3-way interaction analysis and then proceeded to planned comparisons in line with our main questions.

All of our data was analysed by fitting ordinal mixed-effects regressions, using the *ordinal* package in R. The predictors in the models were sum-coded, and both participants and items were included for random effects.² In addition to intercepts, we included random slopes, adopting the following modeling approach to avoid overfitting. We always began by finding the maximal model (in terms of including random slopes) that converged, and then employed model comparison with simpler models to reach a final model. For each random slope in the initial model, we computed a model without it, and then compared the resulting model to the initial maximal model using the *anova* method. A random slope was kept in the final model if it significantly improved model fit (based on the result of the *anova* comparison), and including it led neither to convergence issues (such as non-convergence or

² 'Item' here refers to story-types, where each such type is instantiated by a combination of 'context + test sentence' (see footnote 1).

singularity) nor to high correlations (close to 1 or -1) with other random effects. Note that by-participant random slopes for REDUNDANCY were in general not justified by the design, given its implementation as a between-subjects factor. The details of the model comparison process for each model reported in this paper can be found in the relevant R files on the *OSF* page for this paper, linked [here](#).

2.6 Results

Figure 1 shows the distribution of the responses for the HD and SD conditions, aggregated across ORDER levels and grouped by POLARITY. Overall, we see an effect of REDUNDANCY, in that responses for the positive and negative HD conditions were concentrated toward the lower end of the response scale, around 1-2, whereas responses for the SD baselines descriptively exhibited a different distribution pattern tending towards higher ratings. For the positive cases (SD), responses clustered predominantly around 6-7, indicating high acceptability, whereas for the negative cases (NEGSD), responses were more evenly distributed across the scale, showing greater variability in judgments.

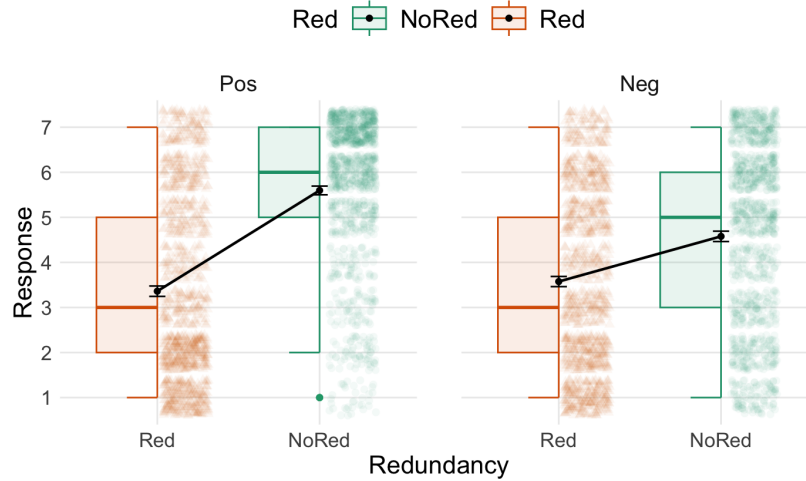


Figure 1 Responses to the HD and SD disjunctions (across ORDER levels) grouped by POLARITY. The data is visualized using a boxplot overlaid with a cloud of individual responses. Black dots represent the mean rating for each condition, with error bars indicating 95% confidence intervals.

We fit an ordinal mixed-effects model (using a *logit* link function) to predict responses based on REDUNDANCY, POLARITY, ORDER and their interactions. Our final model included random intercepts for both participants and items, a by-participant

random slope for POLARITY and by-item random slopes for REDUNDANCY and POLARITY and their interaction. The model revealed significant effects of POLARITY ($\beta = -0.23$, $SE = 0.06$, $z = -4$, $p < .001$) and REDUNDANCY ($\beta = -0.97$, $SE = 0.11$, $z = 8.6$, $p < .001$), as well as a significant POLARITY:REDUNDANCY interaction ($\beta = 0.37$, $SE = 0.04$, $z = -9$, $p < .001$). There were no significant POLARITY:ORDER, ORDER:REDUNDANCY or ORDER:POLARITY:REDUNDANCY interactions. We used the *emmeans* package to conduct planned comparisons with Tukey-adjusted p -values. This analysis revealed significant differences between SD and HD at all levels of combinations of POLARITY and ORDER (positive, first: $\beta = 2.56$, $SE = 0.26$, $z = 9.79$, $p < .0001$; negative, first: $\beta = 1.17$, $SE = 0.24$, $z = 4.91$, $p < .0001$; positive, second: $\beta = 2.82$, $SE = 0.26$, $z = 10.74$, $p < .0001$; negative, second: $\beta = 1.25$, $SE = 0.23$, $z = 5.22$, $p < .0001$). These results indicate that redundancy lowered acceptability for both positive and negative sentences. They also show that the effect of redundancy, while present for both polarities, was stronger for positive sentences, resulting in a significant interaction between REDUNDANCY and POLARITY. This then already provides an answer to Q2 from above: polarity can indeed have a modulating effect on redundancy.

REDUNDANCY and ORDER Turning to Q1, i.e. whether the effect of redundancy is modulated by the order of the disjuncts, we note that we find no ORDER:REDUNDANCY interaction ($\beta = -0.04$, $SE = 0.02$, $z = -1.53$, $p = 0.12$). If ORDER indeed had a modulating effect on REDUNDANCY across the board, we would expect such an interaction. Given that we found that redundancy is modulated by polarity, we might then expect that any potential modulating effect of order on redundancy would be stronger exactly for the level of the POLARITY factor where redundancy is stronger. This possibility predicts the presence of an ORDER:REDUNDANCY interaction for POSITIVE but not NEGATIVE polarities, and consequently should give rise to a three-way POLARITY:ORDER:REDUNDANCY interaction. But the relevant three-way interaction in our model is null ($\beta = 0.02$, $SE = 0.02$, $z = 0.85$, $p = 0.39$).

Given the above, it emerges that we have no evidence for an effect of order on redundancy, at least on the basis of the disjunction data. Of course, we cannot yet conclude that there is no effect of order in disjunctions, given that we currently only have a null result. This question will be addressed more conclusively later in the paper, by comparing different connectives along the dimension of order effects.

2.7 Discussion

Our results indicate that disjunctions with redundant disjuncts are indeed less felicitous than simple disjunctions: Hurford disjunctions, whether positive or negative,

were rated significantly lower than the SD baselines. While this corresponds to the standard reported judgment in the literature, it provides the first experimental confirmation of this effect that we are aware of.

While we do find redundancy effects for both disjunctions with and without negation, the magnitude of the effect is modulated by polarity: it is more pronounced for positive disjunctions than for negative disjunctions. This provides experimental support for the claim that negation impacts oddness judgments for disjunction in Kalomoiros 2023a. The standard approaches to redundancy reviewed above do not account for this modulating role of polarity.

With respect to order, we found no significant effects at the level of the three-way $\text{POLARITY} \times \text{REDUNDANCY} \times \text{ORDER}$ interaction, nor at the level of the lower-order $\text{REDUNDANCY} \times \text{ORDER}$ interactions. To evaluate this lack of evidence for order effects in disjunction more broadly, we will statistically compare the disjunction data to the results for conditionals and conjunctions in Sections 4 and 6, respectively.

3 Experiment 2: Conditionals

Building on the paradigm and design from Experiment 1, Experiment 2 investigated redundancy in conditionals, as well as potential effects of order and polarity.

3.1 Participants

We recruited 195 new participants from Prolific. Participants were screened so that they would only be able to participate in the study if their first language was English; they received monetary compensation labeled ‘Good’ for their participation.

3.2 Design

We tested two types of Hurford conditionals: one with a negated antecedent, and one with a positive antecedent. Since our main link between conditionals and disjunction is the *or-to-if* relation, we code Hurford conditionals with a negative antecedent as *positive* (since they correspond to positive Hurford disjunctions) and Hurford conditionals with a positive antecedent as *negative* (since they correspond to negative Hurford disjunctions).

As in Experiment 1, we manipulated ORDER, in this case by switching the order of antecedent and consequent, resulting in $\text{ANT}(\text{ECEDENT})\text{FIRST}$ vs $\text{ANT}(\text{ECEDENT})\text{SECOND}$ Hurford conditionals. Parallel to the case of disjunction, the FIRST vs SECOND labels for order also encode whether a potential redundancy is introduced in the linearly first or second clause.

- (13) *Negative Hurford Cond* (NEGHC)
 CONTEXT: John and his family want to visit Asia. They have various destinations in mind, but the most prominent ones are cities in Japan, especially Tokyo. Recently, John and his family returned from their Asia trip, so I thought:
 a. If John's family didn't visit Tokyo, they visited Japan. (ANTFIRST)
 b. John's family visited Japan, if they didn't visit Tokyo. (ANTSECOND)
- (14) *Positive Hurford Cond* (PosHC)
 CONTEXT: John and his family want to visit Asia. They have various destinations in mind like some cities in Japan (especially Tokyo), as well as some cities in China. However, I have no idea if they could afford to go to all of these places. Recently, John and his family returned from their Asia trip. I noticed that they made no mention of Tokyo, so I thought:
 a. If John's family visited Japan, they didn't visit Tokyo. (ANTFIRST)
 b. John's family didn't visit Tokyo, if they visited Japan. (ANTSECOND)

As before, we aimed to establish REDUNDANCY effects by comparison to a non-redundant baseline against which to compare the Hurford conditionals:

- (15) *Negative Simple Cond* (NEGSC)
 CONTEXT: John and his family have been wanting to visit Asia for a long time. They considered various options, but in the end had to choose between going to Tokyo in the spring, or visiting some place in China for vacation in the summer. I have no idea what they ended up choosing, but given that the summer has ended now, I know that:
 a. If John's family didn't visit Tokyo, they visited China. (ANTFIRST)
 b. John's family visited China, if they didn't visit Tokyo. (ANTSECOND)
- (16) *Positive Simple Cond* (PosSC)
 CONTEXT: John and his family have been wanting to visit Asia for a long time. They considered various options and initially thought about going both to Tokyo and to some places in China this summer. I have no idea if they had time to do either the Tokyo or the China option. However, I'm sure that they couldn't afford to go to both. So, even though I don't know where they traveled, I know that:
 a. If John's family visited China, they didn't visit Tokyo. (ANTFIRST)
 b. John's family didn't visit Tokyo, if they visited China. (ANTSECOND)

The remainder of the design was identical to Experiment 1 in all relevant respects. A full list of the 24 critical items and 24 fillers used in this experiment is provided in the supplemental materials on OSF.

3.3 Procedure

The procedure was the same as in the previous experiment. The training phase was identical to the one used in Experiment 1 (See Section 2.3 for details).

3.4 Data availability

All stimuli, together with the files and scripts required for the statistical analysis reported below, can be found on the *OSF* page for this paper, linked [here](#).

3.5 Data analysis

The data analysis followed the same approach as that described in the relevant section for the previous experiment.

3.6 Results

Figure 2 illustrates the effects of redundancy across the different combinations of the factor levels of POLARITY and ORDER. The descriptive pattern suggests a 3-way interaction, with a REDUNDANCY $\times$ ORDER interaction present at the NEGATIVE level of the POLARITY-factor, but not at the POSITIVE level.

We assessed this pattern in a 3-way interaction analysis and subsequent planned comparisons via *emmeans* based on our overall questions. Initially, we fit a model predicting response from CONNECTIVE, POLARITY, REDUNDANCY and all their two- and three-way interactions. The final model included random intercepts for participant and item, as well as a by-participant random slope for POLARITY, and a by-item random slope for REDUNDANCY. The model revealed significant main effects of ORDER, POLARITY and REDUNDANCY ($\beta = 0.16$, $SE = 0.02$, $z = 5.86$, $p < .0001$; $\beta = -0.12$, $SE = 0.04$, $z = -2.92$, $p < .001$; $\beta = 0.52$, $SE = 0.1$, $z = 4.8$, $p < .0001$, respectively); of the two-way interactions, POLARITY $\times$ REDUNDANCY was significant ($\beta = 0.13$, $SE = 0.04$, $z = 3.27$, $p < .001$), but ORDER $\times$ POLARITY and ORDER $\times$ REDUNDANCY were not, ($\beta = 0.02$, $SE = 0.02$, $z = 0.84$, $p = 0.39$; $\beta = -0.04$, $SE = 0.02$, $z = -1.72$, $p = 0.08$, respectively). Finally, there was a significant POLARITY $\times$ REDUNDANCY $\times$ ORDER interaction ($\beta = 0.08$, $SE = 0.02$, $z = 3$, $p < .001$).

Because of the the presence of the POLARITY $\times$ REDUNDANCY $\times$ ORDER interaction, we cannot simply look at the POLARITY $\times$ REDUNDANCY and ORDER $\times$ REDUNDANCY interactions in the main model to answer our questions about whether (and when) redundancy is modulated by ORDER and POLARITY; rather, these need to be looked at separately at the different levels of the other modulating factor (ORDER in the case

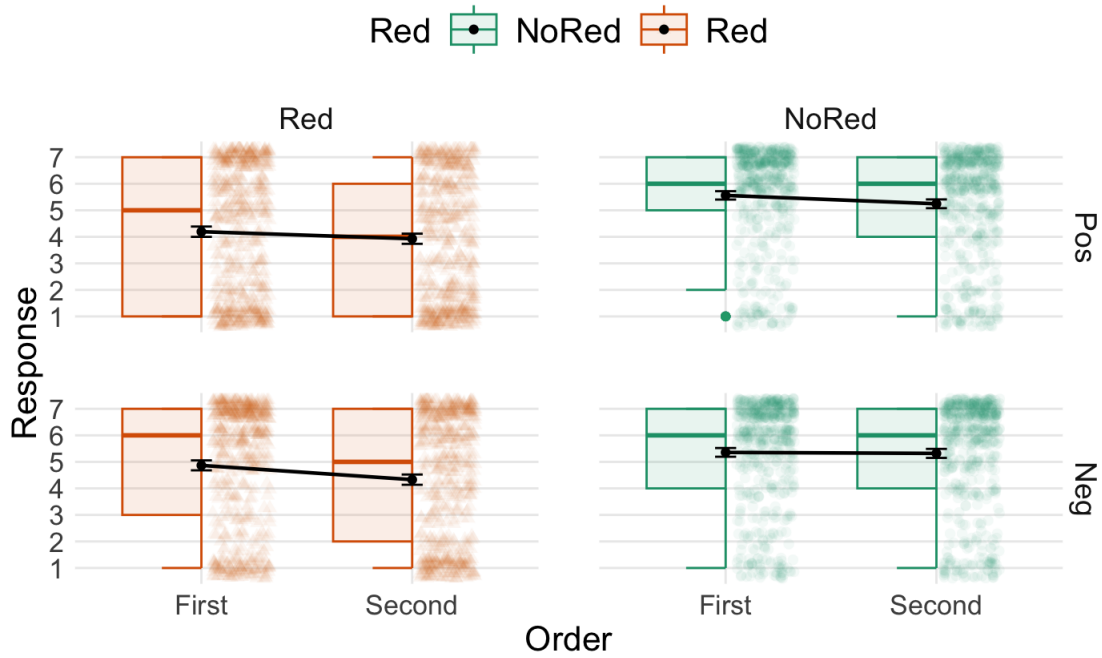


Figure 2 Responses to the HC and SC conditionals (faceted by Polarity) grouped by ORDER. The data is visualized using a boxplot overlaid with a cloud of individual responses. Black dots represent the mean rating for each condition, with error bars indicating 95% confidence intervals.

of $\text{POLARITY} \times \text{REDUNDANCY}$, and POLARITY in the case of $\text{ORDER} \times \text{REDUNDANCY}$). We proceed to do this below.

3.6.1 Polarity $\times$ Redundancy in Hurford conditionals

We start by using *emmeans* to look at the differences between REDUNDANT vs NON-REDUNDANT conditionals at all combinations of levels of the ORDER and POLARITY factors (with Tukey-adjusted p -values). This reveals that redundant conditionals are always significantly different from their non-redundant counterparts, except in the case of antecedent-first conditionals at the NEGATIVE level of polarity.

These estimates also suggest that, at least numerically, POS conditionals are always affected by the REDUNDANCY manipulation more than their NEG counterparts. To test whether this indeed amounts to significant effects, we used *emmeans* again, and extracted the $\text{POLARITY} \times \text{REDUNDANCY}$ interaction at the different levels of the ORDER factor. For ORDER = FIRST, there was a significant interaction, whereby the

Pol	Order	contrast	β	SE	z	p
Neg	First	NoRed - Red	0.52	0.254	2.049	0.4484
Pos	First	NoRed - Red	1.39	0.241	5.777	<.0001
Neg	Second	NoRed - Red	1.045	0.254	4.122	0.0010
Pos	Second	NoRed - Red	1.25	0.240	5.226	<.0001

Table 1 Differences between redundant vs. non-redundant conditionals at all combinations of Order and Polarity Levels. p -values are Tukey-adjusted.

effect of redundancy is different for POS vs. NEG ($\beta = -0.87$, $SE = 0.2$, $z = -4.36$, $p < .0001$); but for ORDER = SECOND, the modulating effect of POLARITY ceases to be significant ($\beta = -0.2$, $SE = 0.19$, $z = -1.04$, $p = 0.29$).

This analysis confirms that polarity indeed plays a role: redundancy effects tend to be stronger for POSITIVE Hurford conditionals (i.e., those with a negative antecedent) compared to their NEGATIVE counterparts (i.e., with positive antecedent). Importantly however, this only gives rise to significant interaction for antecedent-first conditionals; negative antecedent-second conditionals, on the other hand, cannot be statistically distinguished from their positive counterparts in terms of how strongly they are affected by redundancy, even though they are both affected, as is clear from Table 1.

3.6.2 Order $\times$ Redundancy in Hurford conditionals

Turning to order effects, we first extracted the FIRST vs. SECOND contrasts with *emmeans*, at every combination of POLARITY and REDUNDANCY (with Tukey-adjusted p -values). These results suggest that while NEG redundant Hurford conditionals exhibit an order effect that is wholly absent from their non-redundant counterparts, POS Hurford conditionals numerically exhibit an order effects at the NoRED level, and a marginal effect at the RED level.

Next, we extracted the RED:ORDER interactions at both levels of the POLARITY factor. There was a significant ORDER:RED interaction for POL = NEG ($\beta = -0.52$, $SE = 0.15$, $z = -3.31$, $p < .001$), but not for POL = POS ($\beta = 0.14$, $SE = 0.15$, $z = 0.91$, $p = 0.35$), meaning that the three-way POLARITY $\times$ ORDER $\times$ REDUNDANCY interaction is driven by an effect of ORDER ON REDUNDANCY that is polarity-dependent (present for NEG, but absent for Pos).

Pol	Red	Order contrast	β	SE	z	p
Neg	NoRed	First - Second	0.017	0.113	0.150	1.0000
Pos	NoRed	First - Second	0.444	0.112	3.953	0.0020
Neg	Red	First - Second	0.542	0.111	4.900	<.0001
Pos	Red	First - Second	0.302	0.108	2.796	0.0960

Table 2 Differences between Order levels (First vs. Second) at combinations of Polarity and Redundancy. p -values are Tukey-adjusted.

3.7 Discussion

Our results provide experimental evidence for redundancy effects in conditionals. The overall pattern is more complex than in disjunction, as there is a 3-way interaction of REDUNDANCY with POLARITY and ORDER. Our main further findings in this regard can be summarized as follows:

- With the exception of antecedent-first Hurford conditionals at the NEG-level, redundant Hurford conditionals are significantly less natural than their non-redundant baselines, confirming a redundancy effect, at least in three out of the four combinations of the order and polarity factor levels.
- Order and polarity do impact redundancy effects, though their effects are modulated by interactions, in that they only make a difference at certain levels of the relevant other factors.
- In the case of antecedent-first conditionals, the redundancy effect is modulated by POLARITY: it is stronger for POS compared to NEG conditionals (in line with the claims in [Mandelkern & Romoli 2018](#)). On the other hand, we found no evidence for such a polarity-based difference for antecedent-second conditionals. As such, while there was an overall effect of POLARITY on REDUNDANCY (recall the significant POLARITY:REDUNDANCY interaction in our model), this effect is crucially modulated by ORDER.
- We found no effect of ORDER on redundancy (recall the null ORDER:REDUNDANCY in our overall model), which at first blush may make Hurford conditionals appear similar to disjunctions. However, this was again modulated by the POLARITY factor: for NEG conditionals there was a significant ORDER:REDUNDANCY interaction, while for POS conditionals this interaction was null. Therefore, we have an order effect for NEG but not for POS Hurford conditionals; thus, in contrast to disjunctions, order *does* matter for redundancy in conditionals at certain levels of other relevant factors.

A question that arises at this point is how Hurford conditionals stack up against their corresponding disjunctions in direct statistical comparison. This constitutes part of the third question we started with: is the effect of redundancy, and the way this is modulated by other factors like POLARITY and ORDER, constant across connectives? We investigate this in the next section.

4 Cross-connective comparison (Part 1): Disjunction vs. Conditionals

4.1 Data preparation

This section turns to a statistical comparison of disjunction and conditionals. Our aim is to systematically assess whether the impact of the factors modulating redundancy effects differs in any way across connectives.

We created a combined dataset for the Hurford disjunction and the Hurford conditionals data from Experiments 1 and 2. A new CONN(ECTIVE) factor encoded whether a given observation involved conditionals (IF) or disjunction (DISJ). For the POLARITY factor, recall that our guiding intuition for comparisons between Hurford conditionals and Hurford disjunctions was grounded in the *or-to-if* equivalence; this led us to group together HD and negative antecedent conditionals as POS, while NEGHDs and positive antecedent conditionals were grouped together as NEG. The ORDER-factor levels encoded, as before, whether the linearly first or second clause would be redundant in relation to the other clause.

Given the presence of a three-way interaction in conditionals, but no corresponding evidence for disjunctions, we began with a four-way interaction analysis to test whether these three-way interactions differ significantly when compared directly. Thus, we fit a model predicting response from CONNECTIVE, POLARITY, REDUNDANCY, ORDER and all their two-, three- and four-way interactions. The final model included random intercepts for participant and item, as well as by-participant random slopes for REDUNDANCY and ORDER, and by-item random slopes for CONNECTIVE, POLARITY, REDUNDANCY as well as all their two- and three-way interactions.³ The results of the model are summarized in Table 3.

³ Due to the computational difficulties present in fitting a 4-way model, we employed a bottom-up model comparison process. We started with a simple 4-way model that included random intercepts for participant and item, and only included by-item and by-participant random slopes for simple effects. We subsequently attempted adding more random slopes, first testing whether the addition of two-way interaction slopes would improve model fit, and then moving on to three-way interaction slopes (the three-way interaction slopes were only relevant for Item). The models that tested three-way slopes by themselves failed to converge, and as such we did not attempt to include a four-way interaction slope. The model that included both two- and three-way by-item slopes did converge, and as such we decided to include the three-way slopes in the final model out of an abundance of

Term	β	SE	z	p
Conn	-0.350033	0.070557	-4.961	<.0001
Order	0.103748	0.020887	4.967	<.0001
Pol	-0.049444	0.045461	-1.088	0.28
Red	0.728956	0.088080	8.276	<.0001
Conn:Order	-0.081671	0.020799	-3.927	<.0001
Conn:Pol	-0.183680	0.034758	-5.285	<.0001
Order:Pol	-0.028181	0.019299	-1.460	0.14
Conn:Red	0.114855	0.069493	1.653	0.10
Order:Red	-0.050692	0.020915	-2.424	<.05
Pol:Red	-0.245744	0.046325	-5.305	<.0001
Conn:Order:Pol	-0.002087	0.019291	-0.108	0.91
Conn:Order:Red	0.011098	0.020747	0.535	0.59
Conn:Pol:Red	-0.078006	0.032296	-2.415	<.05
Order:Pol:Red	-0.037676	0.019299	-1.952	0.05091
Conn:Pol:Order:Red	0.058314	0.019293	3.023	<.01

Table 3 Result of the model predicting Response from CONNECTIVE, POLARITY, REDUNDANCY, ORDER and all their two-, three- and four-way interactions.

We were interested in knowing: (i) whether different connectives exhibit redundancy effects of differential intensity, (ii) whether the way POLARITY modulates REDUNDANCY differs across connectives, and (iii) whether the way ORDER modulates REDUNDANCY differs across the levels of CONNECTIVE. Importantly, we found a significant four-way $CONNECTIVE \times POLARITY \times ORDER \times REDUNDANCY$ interaction, confirming that the three-way interaction observed for conditionals reflects a statistically distinct pattern from that of disjunctions, where no such interaction was found. Consequently, we cannot restrict our attention to the $CONNECTIVE:REDUNDANCY$, $CONNECTIVE:POLARITY:REDUNDANCY$, and $CONNECTIVE:ORDER:REDUNDANCY$ interactions in the overall model, since the outcomes are expected to vary depending on the levels of the other interacting factors. We therefore examined these interactions separately at each relevant level of the other factors. We begin with the question of whether the effect of REDUNDANCY differs across the two connectives.

caution. The model that excluded these three-way slopes did not differ qualitatively from the final model we report here. More details can be found in the R files on OSF.

4.2 Redundancy effects

To examine whether conditionals vs. disjunctions differ in terms of the strength of the redundancy effects they exhibit, we use *emmeans* to look at the `CONNECTIVE` $\times$ `REDUNDANCY` interactions at each level of the `POLARITY` and `ORDER` factors. The results are summarized in Table 4. The main take-away is that while `NEG` conditionals and disjunctions do not show a significant contrast in either order, `POS` conditionals and disjunctions do show differences in terms of how they are affected by the `RED` vs. `NORED` contrasts: a marginal effect for `ORDER = FIRST` and a clearly significant effect for `ORDER = SECOND`.

Pol	Order	Conn contrast	Red contrast	β	SE	z	p
Neg	First	Or - If	NoRed - Red	0.425	0.334	1.271	0.20
Pos	First	Or - If	NoRed - Red	0.583	0.311	1.875	0.06
Neg	Second	Or - If	NoRed - Red	-0.130	0.336	-0.387	0.69
Pos	Second	Or - If	NoRed - Red	0.960	0.325	2.955	<.01

Table 4 Conn:Red Interaction Contrasts at Combinations of Polarity and Order

A useful way to interpret these results is as follows: the polarity factor groups together conditionals and disjunctions that are equivalent under the *or-to-if* inference. As shown above, `NEG`HDs and conditionals at the `NEGATIVE` level of `POLARITY` (with a positive antecedent) exhibit parallel effects of redundancy. In contrast, HDs and conditionals at the `POSITIVE` level (with negative antecedents) display different redundancy effects in the `SECOND` order, and marginally different effects in the `FIRST` order. This pattern is illustrated in Fig. 3. Notably, the difference between Hurford conditionals and their corresponding disjunctions is most pronounced in the `POL = POS` case, as reflected by the steeper slope for disjunctions in the left two subplots.

We used *emmeans* to pull out the individual `RED` vs. `NORED` contrasts for each connective, at each level of the `ORDER` and `POLARITY` factors. The results, summarized in Table 5, suggest that the intensity of redundancy effects is stronger for positive HDs compared to their corresponding positive HCs. It also interesting to note that `NEGFIRST` disjunctions come with a significant redundancy effect ($p = 0.0006$), whereas the corresponding `NEGFIRST` conditionals do not ($p = 0.1634$). While this difference does not give rise to a significant `CONN:RED` interaction at the `NEGFIRST` level (see table 4), it still suggests that the relevant redundancy effect is more pronounced for `NEGFIRST` disjunctions than `NEGFIRST` conditionals.

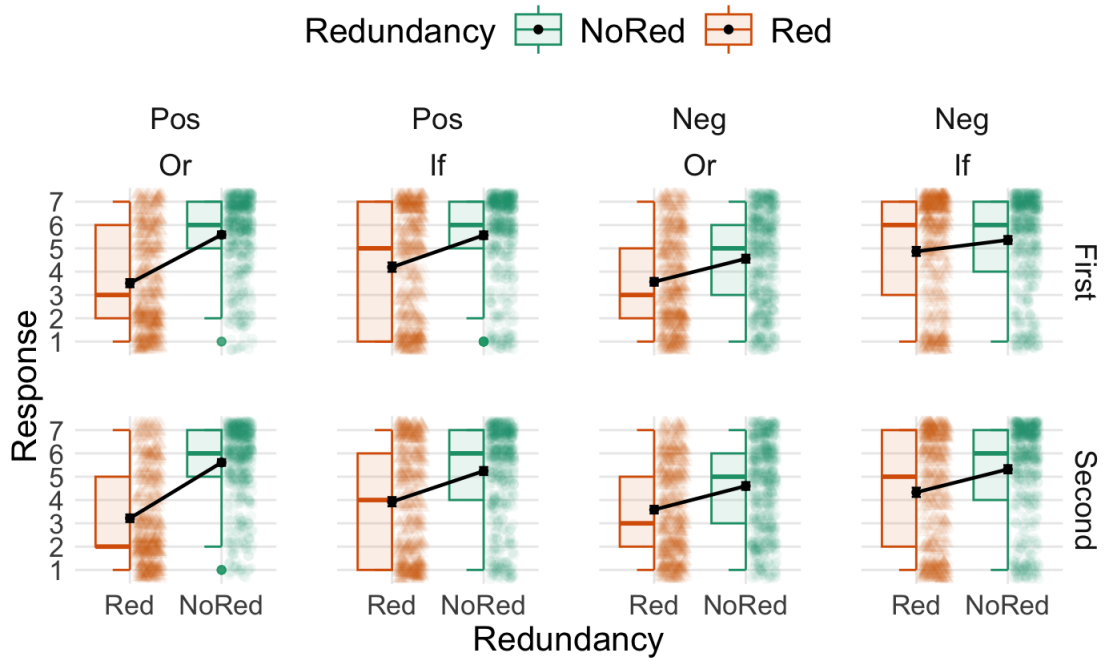


Figure 3 Responses to the NoRED vs RED disjunctions, contrasted to their corresponding conditionals, by Order and Polarity. The data is visualized using a boxplot overlaid with a cloud of individual responses. Black dots represent the mean rating for each condition, with error bars indicating 95% confidence intervals.

Pol	Order	contrast	β	SE	z	p
Neg	First	Or NoRed - Or Red	1.002	0.258	3.878	0.0006
Neg	First	If NoRed - If Red	0.577	0.279	2.069	0.1634
Pos	First	Or NoRed - Or Red	2.215	0.247	8.976	<.0001
Pos	First	If NoRed - If Red	1.632	0.260	6.286	<.0001
Neg	Second	Or NoRed - Or Red	1.078	0.260	4.150	0.0002
Neg	Second	If NoRed - If Red	1.208	0.280	4.309	0.0001
Pos	Second	Or NoRed - Or Red	2.456	0.257	9.564	<.0001
Pos	Second	If NoRed - If Red	1.495	0.267	5.592	<.0001

Table 5 NoRed vs. Red contrasts for OR vs. IF, at combinations of POLARITY and ORDER. p values are Tukey-adjusted.

4.3 Effects of Polarity on Redundancy

To examine whether POLARITY affects REDUNDANCY differently across the two connectives, we use *emmeans* to pull out the CONNECTIVE $\times$ POLARITY $\times$ REDUNDANCY interaction at both levels of the ORDER factor (see Table 6).

Order	Conn	Pol	Redundancy	β	SE	z	p
First	Or - If	Neg - Pos	NoRed - Red	-0.158	0.302	-0.522	0.60
Second	Or - If	Neg - Pos	NoRed - Red	-1.091	0.300	-3.632	<.001

Table 6 Three-way interaction contrasts (Conn:Pol:Red) at each Order level

For ORDER = FIRST, the effect of polarity on redundancy does not differ significantly across the two connectives. In contrast, a significant three-way interaction emerged for ORDER = SECOND. This pattern is further illustrated in Fig. 4. Impressionistically, this appears to reflect that, for ORDER = SECOND, the NORED vs RED contrast is larger for OR than for IF. This impression is confirmed using *emmeans* to extract the POLARITY:REDUNDANCY interactions at each level of CONN and ORDER (see Table 7). Specifically, for ORDER = FIRST, both connectives show parallel POLARITY:REDUNDANCY interactions. However, for ORDER = SECOND, the interaction is stronger for OR than for IF.

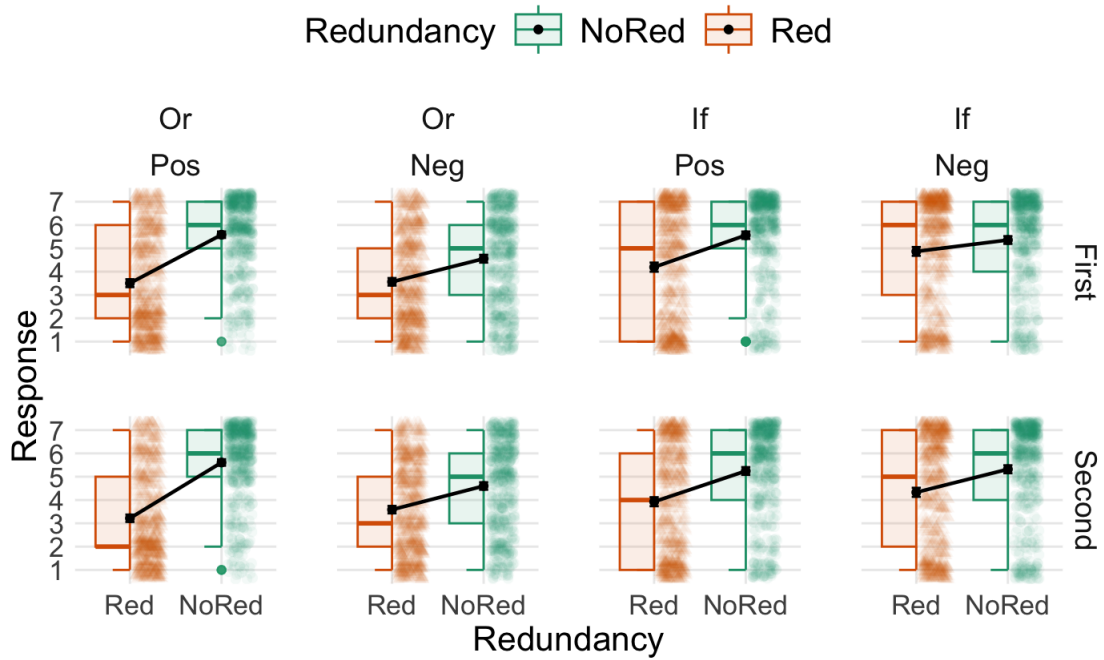


Figure 4 Responses to the NoRED vs RED conditions, contrasted by Polarity, and faceted Order and Connective. The data is visualized using a boxplot overlaid with a cloud of individual responses. Black dots represent the mean rating for each condition, with error bars indicating 95% confidence intervals.

Conn	Order	Pol	Red	β	SE	z	p
Or	First	Neg - Pos	NoRed - Red	-1.212	0.251	-4.834	<.0001
If	First	Neg - Pos	NoRed - Red	-1.055	0.252	-4.188	<.0001
Or	Second	Neg - Pos	NoRed - Red	-1.378	0.251	-5.489	<.0001
If	Second	Neg - Pos	NoRed - Red	-0.287	0.250	-1.149	0.2506

Table 7 Pol:Red interaction contrasts at combinations of Connective and Order

In summary, polarity modulates redundancy for both connectives. However, this modulation differ across these connectives depending on ORDER: for ORDER = FIRST, the pattern is parallel, but for ORDER = SECOND, polarity significantly influences redundancy for OR but not for IF.

4.4 Effects of Order on Redundancy

Finally, we consider how the ORDER:REDUNDANCY interaction differs between the two connectives. For conditionals, we have seen that there was an ORDER:REDUNDANCY interaction further modulated by POLARITY, yielding a three-way interaction. In contrast, disjunctions showed neither an ORDER:REDUNDANCY interaction nor a POLARITY:ORDER:REDUNDANCY interaction. The four-way model further supports this picture, with the previously reported significant four-way interaction confirming that this dimension varies across connectives. Analyses of the (No)REDFIRST vs. (No)REDSECOND contrasts at each level of CONNECTIVE and POLARITY (Table 8) reinforce these conclusions: NEG conditionals exhibit an order effect absent in their POS counterparts, while neither POS nor NEG disjunctions show any order effect. These polarity-mediated order differences across the two connectives are further illustrated in Fig. 5.

Conn	Pol	contrast	β	SE	z	p
Or	Neg	First NoRed - Second NoRed	-0.05430	0.108	-0.503	0.9583
Or	Neg	First Red - Second Red	0.02153	0.104	0.207	0.9969
Or	Pos	First NoRed - Second NoRed	-0.01578	0.114	-0.139	0.9991
Or	Pos	First Red - Second Red	0.22515	0.105	2.152	0.1368
If	Neg	First NoRed - Second NoRed	0.00309	0.121	0.026	1.0000
If	Neg	First Red - Second Red	0.63421	0.119	5.342	<.0001
If	Pos	First NoRed - Second NoRed	0.49143	0.121	4.061	0.0003
If	Pos	First Red - Second Red	0.35463	0.117	3.042	0.0126

Table 8 Selected Order contrasts (First vs. Second) at combinations of Connective and Polarity. p values are Tukey-adjusted.

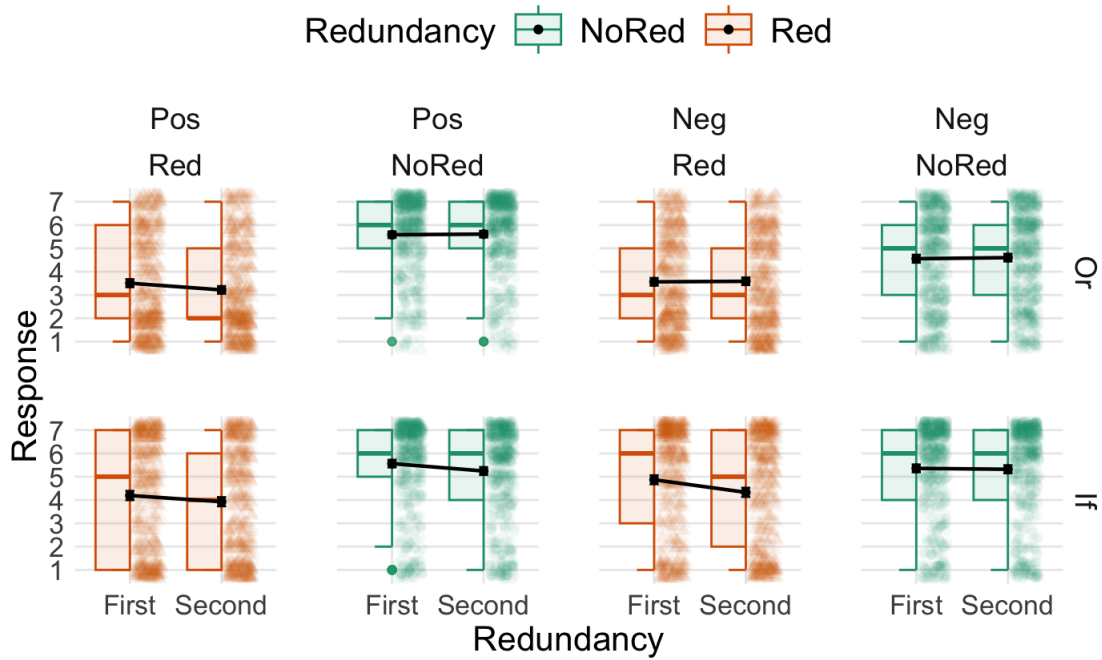


Figure 5 Responses to REDFIRST vs REDSECOND conditions, contrasted to their corresponding NoRED conditions, and faceted by Order and Connective. The data is visualized using a boxplot overlaid with a cloud of individual responses. Black dots represent the mean rating for each condition, with error bars indicating 95% confidence intervals.

4.5 Interim summary

We are faced with a complex landscape: overall, the two connectives differ across all three dimensions of interest, but these differences are further shaped by the presence of interactions:

- Disjunction and conditionals exhibit differential effects of redundancy, but this difference is only observed for POS polarities, where disjunctions show a stronger effect of redundancy than conditionals.⁴ For NEG polarities, the effects of redundancy appear statistically parallel.

⁴ As noted earlier, for ORDER = FIRST, the difference in the strength of redundancy between the two connectives is marginal ($p = .06$). For simplicity, we describe this contrast as occurring in the Pos polarity, since $p = .06$ is near the conventional threshold for significance, and doing so avoids over-complicating an already complex picture.

- Polarity affects redundancy differently across the two connectives, but the difference is only observed for the `SECOND` order: the redundancy effect in disjunctions is stronger for `Pos` than for `NEG` whereas, in the corresponding conditionals, polarity doesn't significantly affect the strength of redundancy. On the other hand, for `FIRST`, `Pos` polarity leads to a stronger redundancy effect than `NEG`, for both connectives.
- Order modulates redundancy differently across the two connectives. Conditionals show a stronger `ORDER` effect on `REDUNDANCY` than disjunctions; however, this effect is observed only for the `NEG` polarity and is absent for `POS`.

In summary, disjunctions and conditionals share some common patterns: in both, switching from `Pos` to `NEG` polarity reduces the strength of redundancy, and overall there is no significant effect of order on redundancy. Crucially, however, while these patterns hold consistently for disjunctions across levels of `POLARITY` and `ORDER`, for conditionals they are modulated by interactions: the ameliorating effect of `NEG` polarity occurs only for `FIRST` order, and the absence of an order effect is observed only for `Pos` polarity.

These findings make it difficult to treat the two connectives on par. Setting aside order effects for now, we could, as a first pass, interpret the redundancy effects in disjunctions as follows: `NEG HD` disjunctions, being worse than their `SD` baselines, involve one type of redundancy violation. `Pos HDs`, which are degraded even more relative to their `SD` baselines, carry two types of redundancy violations: the one shared with their `NEG` counterparts, plus an additional violation specific to them. An example of how such a theory can be implemented is provided in Section 7.

Now, if we assume that conditionals should behave like their *or-to-if* equivalent disjunctions in terms of the strength of redundancy, then we would expect `NEG` conditionals to always be less redundant than their `Pos` counterparts, and to not contrast with disjunctions. But our results show that, at least for `ORDER = SECOND`, conditionals show no effect of polarity, whereas for `ORDER = FIRST`, they do show such an effect. This leads to a significant `CONNECTIVE × POLARITY × REDUNDANCY` interaction on the `SECOND` level of the `ORDER` factor, which is unexpected under a theory of complete parallelism between the two connectives. In addition, note that, even for the `FIRST` order, where the two connectives pattern more closely together, it's suspicious that we see no significant redundancy effect for `NEG` conditionals, but do see one for `NEG` disjunctions (even if this doesn't lead to a significant `CONNECTIVE × REDUNDANCY` interaction). The expectation is that there should be such a significant redundancy effect for `NEGFIRST` conditionals.

A possibility then to consider is that, while two redundancy effects might be at play in disjunctions, only one is at play in conditionals: NEGFIRST conditionals are not redundant, whereas POSFIRST conditionals are. In that respect, FIRST conditionals mirror disjunctions, where switching from NEG to POS adds a redundancy violation. But for SECOND conditionals both POS and NEG carry just the one redundancy violation and are thus equally redundant, in contrast to disjunctions. We will return to this issue in the general discussion. For now, we turn to data on another connective, conjunction, to gain a more complete understanding of how redundancy effects vary across connectives.

5 Experiment 3: Conjunction

The third experiment explores redundancy and order effects in conjunction.

5.1 Participants

A total of 201 participants were recruited from our university's Psychology subject pool, 195 of whom had not participated in Experiment 1. There was no overlap with participants in Experiment 2. All participants were undergraduate native English speakers.

5.2 Design

The design of this experiment was based on Experiments 1 and 2 (see Section 2.2), with one key difference: we did not include a polarity manipulation. According to prior literature, conjunction is a connective where order effects may be expected, whereas no analogous claims have been made regarding polarity. By homing in on a connective with potential general order effects, we aimed to gain further perspective on the absence of order effects for other connectives (at least at certain levels of our factors, in the case of conditionals), once we turn to further direct cross-connective comparisons in the next section. Therefore, our critical stimuli consisted solely of positive redundant conjunctions in either order, which we refer to as Hurford Conjunctions (HCONJ).

(17) *Hurford Conjunction* (HCONJ)

CONTEXT: John and his family want to visit Asia. They have various destinations in mind, but the most prominent ones are cities in Japan, especially Tokyo. Recently, John and his family returned from their Asia trip, and had various souvenirs that I suspect they got in Tokyo (although I'm not entirely sure). So, I thought:

- a. John’s family visited Japan and they visited Tokyo. (FIRST)
- b. John’s family visited Tokyo and they visited Japan. (SECOND)

To assess effects of REDUNDANCY, we again include baselines with no violation or redundancy, similar to Experiments 1 and 2.

(18) *Simple Conjunction* (SCONJ)

CONTEXT: John and his family have been wanting to visit Asia for a long time. They recently came into a bit of money, and were talking about traveling to Tokyo and to China. They were also debating going to Australia, but not everyone was up for that. While I don’t know what their final itinerary entailed, I suspect at least that:

- a. John’s family visited China and they visited Tokyo. (FIRST)
- b. John’s family visited Tokyo and they visited China. (SECOND)

The rest of the design was identical to that of the previous experiments in all relevant respects. As before, a total of 24 critical items were created, along with 24 fillers, which are omitted here for brevity.

5.3 Procedure

The procedure was the same as in Experiment 1 (see Section 2.3), except for the training phase, which had a different training trial. Instead of redundant vs non-redundant conjunctions, participants saw the following three training trials: (i) a simple sentence in a context, with no violations of any kind, (ii) a sentence that was contradictory in its context, and (iii) a sentence that contained an implicature violation relative to its context.

5.4 Data availability

All stimuli, together with the files and scripts required for the statistical analysis reported below, can be found on the OSF page for this paper, linked [here](#).

5.5 Data analysis

The data analysis followed the same approach as that described in Section 2.5 for Experiment 1. Our key questions were whether there is an effect of REDUNDANCY and whether such effects are modulated by ORDER, as would be reflected in a REDUNDANCY $\times$ ORDER interaction.

5.6 Results

Figure 6 shows the distribution of the responses for the redundant and non-redundant conjunctions by ORDER.

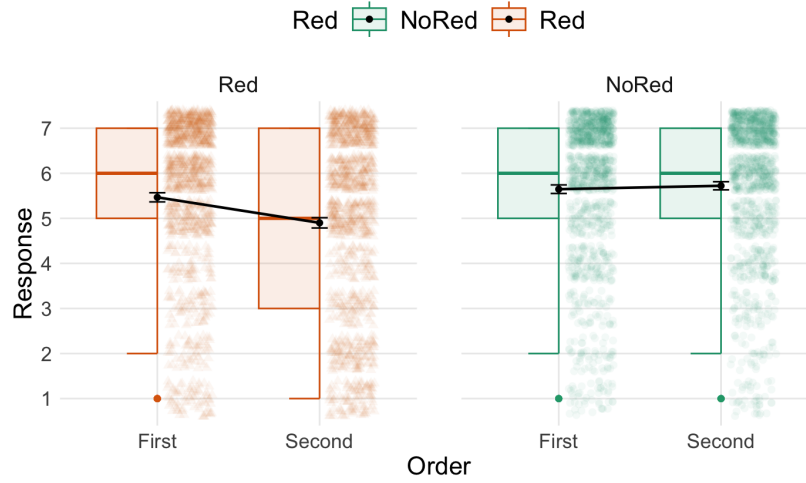


Figure 6 Responses to the Hurford and Simple conjunctions by ORDER. The data is visualized using a boxplot overlaid with a cloud of individual responses. Black dots represent the mean rating for each condition, with error bars indicating 95% confidence intervals.

We fit a model predicting Response from REDUNDANCY, ORDER and their interaction. The final model included random intercepts for participant and item, as well as a by-item random slope for REDUNDANCY. The model revealed significant main effects of ORDER ($\beta = 0.16$, $SE = 0.02$, $z = 5.85$, $p < .001$) and REDUNDANCY ($\beta = -0.38$, $SE = 0.13$, $z = -2.8$, $p < .01$), as well as a significant interaction between these two factors ($\beta = 0.17$, $SE = 0.02$, $z = 6.36$, $p < .001$). It is clear that this interaction is driven by an order effect that is present for RED but absent in the NoRED baseline. Pulling out the relevant contrasts with *emmeans* (Tukey-adjusted values), we see a significant difference for REDFIRST vs. REDSECOND ($\beta = 0.68$, $SE = 0.07$, $z = 8.8$, $p < .0001$), but no such difference for NoREDFIRST vs. NoREDSSECOND ($\beta = -0.02$, $SE = 0.08$, $z = -0.34$, $p = 0.98$).

5.7 Discussion

The results of Experiment 3 confirm the standard claim that $p^+ \wedge p$ is infelicitous. They also reveal an interaction between REDUNDANCY and ORDER for conjunctions, driven by an order effect at the RED-level, whereas no such effect was found for

the control NORED conjunctions. This experimentally supports prior reports, based on informal judgments, that order modulates redundancy effects in conjunctions. Importantly, this contrasts with our findings for disjunctions and conditionals (at least for the Pos polarity), highlighting a theoretical challenge: standard accounts of redundancy generally predict uniform (a-)symmetries across connectives, yet our experiments reveal a more complex pattern.

We now turn to a cross-connective comparison of conjunction with disjunctions and conditionals, analyzing pooled data for the relevant pairings.

6 Cross-connective comparison (Part 2): Conjunction vs. Disjunction and Conditionals

This section presents statistical comparisons of conjunction with disjunctions and conditionals, pooling data from the respective sub-experiments. Our goal is to systematically test whether these connectives differ in their redundancy and order-based effects.

6.1 Data preparation

Comparing the effects of redundancy and order in conjunction with those in the other two connectives required combining different parts of our original datasets from Experiments 1, 2, and 3. In particular, since we did not manipulate polarity for conjunction in Experiment 3, and since polarity played an important role in disjunction and conditionals, we will only compare the conjunction data to what seem like the most comparable polarity-variants. For disjunction, this is the HD-variant, with two positive disjuncts and no negation. By extension, for conditionals, we look at the variant that stands in the *or-to-if*-inference relation to HD, that is, the NEGANT-conditionals. We analyzed the two pairings of connective separately in the corresponding pooled datasets. For each dataset, we then coded a CONN(ECTIVE) factor to indicate whether each observation involved IF, DISJ, or CONJ, based on its source dataset. The coding for ORDER and REDUNDANCY factors was the same as before.

6.2 Conjunction vs. Disjunction

A visual comparison of Figure 6 (conjunction) and Figure 1 (disjunction) suggests differences in whether REDUNDANCY $\times$ ORDER interact. To assess this statistically, we fit a model predicting responses from CONNECTIVE, REDUNDANCY, ORDER and their interactions. The final model included random intercepts for participant and items,

as well as by-item random slopes for CONNECTIVE, REDUNDANCY, and the CONNECTIVE $\times$ REDUNDANCY interaction. Additionally, a by-participant random slope for ORDER was included. The model revealed a significant three-way interaction ($\beta = -0.06$, $SE = 0.02$, $z = -2.18$, $p < .05$). Planned comparisons using *emmeans*, indicated that this interaction was driven by a REDUNDANCY:ORDER interaction for conjunctions ($\beta = -0.8$, $SE = 0.14$, $z = -5.68$, $p < .0001$), which, recall, was not observed for disjunction ($\beta = -0.31$, $SE = 0.17$, $z = -1.77$, $p = 0.07$).

Beyond order, the results also indicate that while both conjunctions and disjunctions exhibit redundancy effects, the impact is significantly stronger for disjunctions. Figure 7 visualizes this contrast, plotting positive Hurford disjunctions against their SD baselines and comparing them to conjunctions (also plotted against their SC baselines).

To assess this statistically, we carried out planned comparisons that tested whether the difference between RED vs NoRED conditions differed between disjunction and conjunction. We evaluated this comparison at the SECOND level of the ORDER factor for conjunction (since in that case we know that conjunctions show an effect of redundancy – no effect of redundancy is found in conjunction for ORDER = FIRST.⁵); but since disjunction exhibited no order effect, we averaged over the level of the ORDER factor for disjunction. The result was that indeed the two connectives differ in how strong a redundancy effect they exhibit, leading to significant interaction ($\beta = 2.41$, $SE = 0.36$, $z = 6.56$, $p < .0001$).

6.3 Discussion

We contrasted the conjunction results from Experiment 3 with the disjunction data from Experiment 1, yielding two main conclusions. First, the two connectives differ in how order affects redundancy: conjunctions show a strong order effect, whereas disjunctions do not, resulting in a significant CONNECTIVE $\times$ ORDER $\times$ REDUNDANCY interaction. This further supports the absence of an order effect for disjunctions. Second, redundancy effects differ by connective: disjunctions exhibit stronger redundancy effects than conjunctions.

6.4 Conjunction vs. Conditionals

To assess how redundancy effects in conditionals relate to redundancy effects in conjunctions, we subsetting our data to include only Pos conditionals and con-

⁵ Using *emmeans* on the model of section 5 to extract the RED vs NoRED at ORDER = FIRST we get: $\beta = 0.42$, $SE = 0.28$, $z = 1.48$, $p = 0.44$

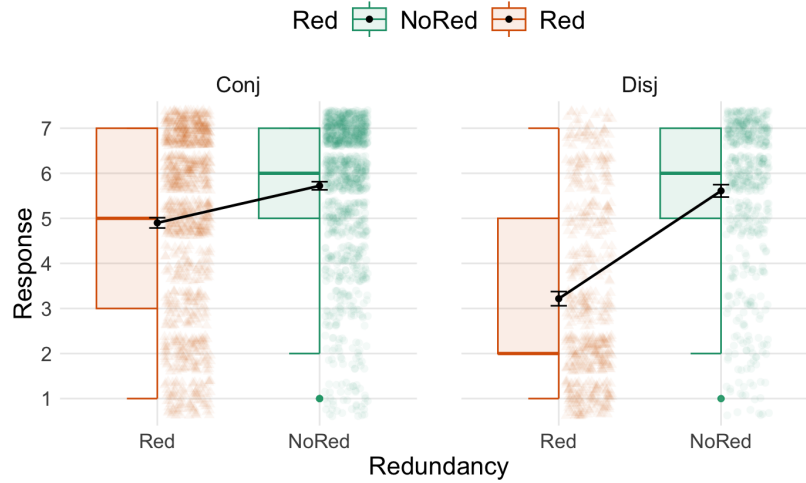


Figure 7 Responses to Red vs NoRed grouped by CONNECTIVE. Both connectives are visualized only at the Second level of the ORDER factor. The data is visualized using a boxplot overlaid with a cloud of individual responses. Black dots represent the mean rating for each condition, with error bars indicating 95% confidence intervals.

junctions.⁶ As before, we were interested in knowing whether the strength of redundancy differ between the two connectives, and whether the two connectives behave differently in terms of the modulating effect of order on redundancy.

To answer the questions, we fit a model predicting Response from CONNECTIVE, REDUNDANCY, ORDER and all their two- and three-way interaction. The final model included random intercepts for participant and item, as well as a by-participant random slope for ORDER, and by-item slopes for CONNECTIVE, REDUNDANCY, ORDER, and CONNECTIVE:REDUNDANCY. The model revealed significant main effects of CONNECTIVE, REDUNDANCY and ORDER (CONN: $\beta = 0.38$, $SE = 0.09$, $z = 4.01$, $p < .001$, REDUNDANCY: $\beta = 0.58$, $SE = 0.09$, $z = 6.09$, $p < .001$, ORDER: $\beta = 0.18$, $SE = 0.03$, $z = 5.34$, $p < .001$). There were also two significant two-way interactions: CONNECTIVE:REDUNDANCY ($\beta = -0.22$, $SE = 0.09$, $z = -2.36$, $p < .05$), and REDUNDANCY:ORDER ($\beta = -0.07$, $SE = 0.03$, $z = -2.34$, $p < .05$); the CONNECTIVE:ORDER interaction did not rise to significance ($\beta = -0.04$, $SE = 0.03$, $z = 1.44$, $p = 0.14$). Finally, the CONNECTIVE:REDUNDANCY:ORDER interaction was significant ($\beta = -0.1$, $SE = 0.03$, $z = -3.56$, $p < .001$)

⁶ Recall our reasoning: we aim to compare conjunctions to the conditionals most similar to them. Since conjunctions were previously compared to Pos disjunctions (as they lack a polarity manipulation), we now compare them to the conditionals equivalent to Pos disjunctions under the *or-to-if* mapping—namely, Pos conditionals.

With this in mind, we ask whether the effect of order on redundancy differs between the two connectives. The answer is emphatically ‘yes’, as indicated by the significant $\text{CONNECTIVE} \times \text{REDUNDANCY} \times \text{ORDER}$ interaction. Figure 8 illustrates this clearly: conjunctions show an order effect that is absent for Pos conditionals. Planned comparisons in Table 9 further confirm this pattern.

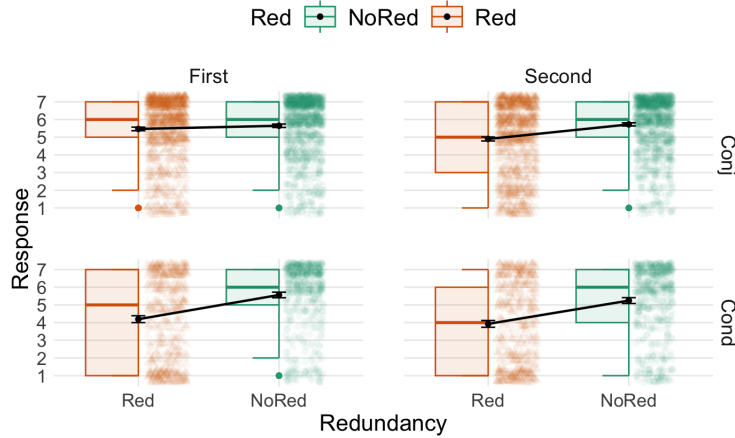


Figure 8 Responses to REDUNDANT vs SIMPLE conjunctions and conditionals, faceted by ORDER. The data is visualized using a boxplot overlaid with a cloud of individual responses. Black dots represent the mean rating for each condition, with error bars indicating 95% confidence intervals.

Conn	Red	Order	β	SE	z	p
And	NoRed - Red	First - Second	-0.73	0.15	-4.81	<.0001
If	NoRed - Red	First - Second	0.14	0.19	0.76	0.44

Table 9 Red:Order interactions at each level of Connective

Moving on to the question of whether the strength of redundancy differs between the two connectives, the significant $\text{CONNECTIVE} : \text{REDUNDANCY}$ interaction would indicate a positive answer to this. However, recall that, in our data, the effect of redundancy in conjunctions is only truly present for SECOND conjunctions, whereas for Pos conditionals it was present for both orders. Therefore, we used *emmeans* to extract a custom contrast comparing NoRED vs. RED across the two connectives in the following manner: the NoRED vs. RED contrast was computed averaged across both order levels for IF, but only on the SECOND order level for AND. This contrast is depicted in Fig. 9 and summarized in Table 10. While it looks like the effect is stronger for conditionals than conjunctions, this difference turns out to be non-significant.

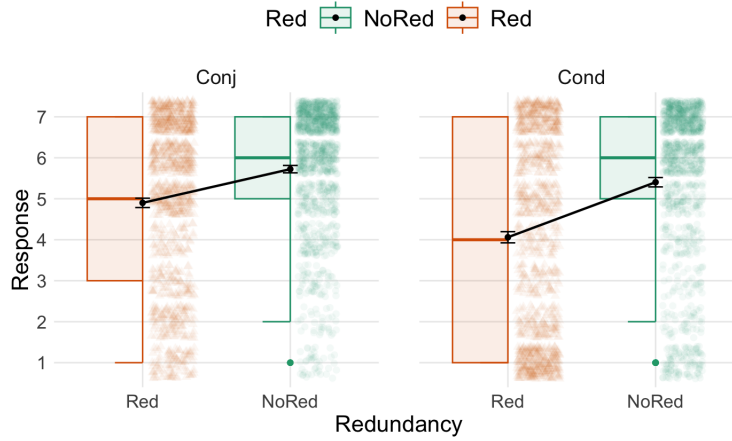


Figure 9 Responses to the Pos conditionals (REDUNDANT vs SIMPLE), averaged across ORDER levels, and contrasted with responses to SECOND conjunctions. The data is visualized using a boxplot overlaid with a cloud of individual responses. Black dots represent the mean rating for each condition, with error bars indicating 95% confidence intervals.

Conn	Red	β	SE	z	p
And - If	NoRed - Red	-0.51	0.38	-1.33	0.18

Table 10 Pairwise interaction contrast for CONNECTIVE and REDUNDANCY

6.5 Discussion

When comparing conjunctions to disjunctions, we found that these connectives differ in terms of both order effects and strength of redundancy. The comparison of conjunctions to the relevant conditionals corresponding to Pos disjunctions upheld these results only for order effects: conjunctions show an order effect that is absent for Pos conditionals, just as it was absent for disjunctions. However, the strength of redundancy isn't statistically distinguishable between conjunctions and conditionals, whereas Pos disjunctions showed stronger effects of redundancy than conjunctions. This follows a motif that we have already encountered: Pos conditionals aren't as extreme in the strength of their redundancy violation as Pos disjunctions.

We now turn to a more detailed discussion of the effects identified across our three experiments, aiming to explore how current theoretical frameworks might be adapted to accommodate them.

7 General discussion

7.1 Summary of the findings

Our first contribution consists in providing, to our knowledge, the first experimental *vindication* of the core intuitions driving standard approaches to redundancy: namely, that Hurford disjunctions, redundant conjunctions, and Hurford conditionals are degraded relative to non-redundant baselines.

Our second main contribution is the finding that redundancy is less of a binary phenomenon than previously assumed; instead, it can be modulated by factors such as polarity, order and the identity of the connective. At a high level, these modulating effects can be described as follows:

- Differential effects of order across connectives.
- Polarity sensitivity of redundancy judgments.
- Differential strength of redundancy effects across connectives.

As we have emphasized throughout this paper, this general picture is further shaped by additional interactions. Yet even at this general descriptive level, it is evident that these patterns are not predicted by standard versions of the Simplification and Local Contexts theories of redundancy outlined at the outset. First, these theories are most naturally implemented to predict *either* uniform asymmetries *or* uniform symmetries across connectives. Second, they rely on notions of (contextual) entailment that are insensitive to the presence or absence of negation, as long as the entailment relation is held fixed. Finally, these theories produce binary judgments for a sentence and lack the resources to account for differential effects of redundancy across connectives.

At this stage, it is helpful to synthesize the patterns revealed in our experiments that any theory aiming to account for our results will need to accommodate:

- i. Both POSITIVE and NEGATIVE Hurford disjunctions exhibit redundancy effects; but the relevant effect is stronger for the POSITIVE cases.
- ii. Redundancy effects for disjunction are not modulated by ORDER.
- iii. Redundancy effects for conjunction are modulated by ORDER: the SECOND order (i.e., $p^+ \wedge p$) exhibits effects of redundancy, but the FIRST order (i.e., $p \wedge p^+$) does not.
- iv. The redundancy effect observed with the SECOND order in conjunctions is weaker than that observed in POSITIVE disjunctions.

- v. NEGFIRST conditionals show no effect of redundancy, but NEGSECOND ones do. POSFIRST and POSSECOND conditionals show comparable redundancy effects. As such, conditionals exhibit a polarity-dependent order effect.
- vi. POSITIVE conditionals are less redundant than POSITIVE disjunctions.
- vii. The strength of redundancy in POSITIVE conditionals does not significantly differ from that of SECOND conjunctions.

In what follows, we will suggest possible directions for capturing these effects. Specifically, we will suggest order effects might profitably be derived from a theory that more closely integrates redundancy and presupposition. Effects that have to do with the differential strength of redundancy between connectives will be examined via the lens of the theory of *super-redundancy*: this approach can help account for the relevant ‘strength’ and polarity effects that we see in conjunctions and disjunctions, but does not straightforwardly extend to conditionals, which are then left as a puzzle.

7.2 Order effects and presupposition

We start with the fact that different connectives show different sensitivity to order effects, and specifically with the fact that these effects are present in conjunction but not disjunction. Interestingly, a similar pattern of order effects has emerged in recent work on presupposition projection. It has long been argued, since at least Karttunen (1974), that presupposition filtering is sensitive to order in conjunctions, but not in disjunctions. For example, consider the following:

- (19) a. John used to smoke Marlboros and he stopped.
- b. John stopped smoking and he used to smoke Marlboros.

While (19a) is not felt to presuppose that John used to smoke, (19b) is interpreted as presupposing this, a conclusion experimentally confirmed by Mandelkern et al. (2020).⁷ By contrast, in disjunctions like the examples below, no presupposition is felt to arise, regardless of order (see Kalomoiros & Schwarz 2024 for recent experimental confirmation).⁸ This demonstrates that, unlike conjunction, disjunction exhibits symmetric behavior with respect to presupposition filtering.

⁷ In case where one conjunct is equivalent to the presupposition of the other (e.g., *John used to smoke and he stopped*), one might be tempted to attribute the order effects to redundancy/triviality. Adding ‘Marlboros’ to make the relevant relation one of asymmetric entailment controls for this potential confound. The results in Mandelkern et al. (2020) indicate that, even with this adjustment, the contrast persists.

⁸ One issue that must be controlled for when reasoning about symmetric filtering with disjunctions is the possibility that symmetry arises from local accommodation (Schlenker 2008, Hirsch & Hackl

- (20) a. Either John didn't use to smoke or he stopped.
 b. Either John stopped smoking or he didn't use to smoke.

The above patterns suggest a parallelism between order effects in presupposition projection and order effects in redundancy, concordant with the idea that positions where a presupposition can be filtered are positions where it has somehow become known information (Stalnaker 1974, Schlenker 2009: a.o.). In the domain of filtering, various theoretical frameworks have been developed to try to account in an explanatory way for the differential order effects between conjunction vs disjunction (George 2008, 2007, Kalomoiros 2023b, Kalomoiros & Schwarz 2024). Nevertheless, such frameworks have not yet been extended to cover the redundancy side. In fact, it remains to be seen how far the parallelism between filtering and redundancy goes, and how these frameworks can be best transferred to cover the results on redundancy we have presented in the current paper.

From the perspective of our results, a core question along the lines of the above mentioned parallelism, is whether order effects in filtering with conditionals follow the same pattern as their redundant Hurford counterparts. Recent work suggests that this might be the case. In particular, Kalomoiros et al. (2024) present experimental results suggesting that conditionals with *negated* antecedents exhibit symmetric filtering. Specifically, their results show that the presupposition of the antecedent is filtered in both (21a) and (21b):

- (21) a. If John isn't studying in Japan too, Mary is not studying in Tokyo.
 b. Mary is not studying in Tokyo, if John isn't studying in Japan too.

This mirrors the absence of order effects in POSITIVE Hurford conditionals (whose antecedents were negated). Pushing this line of reasoning, we expect that, given the order effect we found with NEGATIVE Hurford conditionals, a similar effect should be found on the filtering side once we remove the negation from the antecedents in (22). Specifically, we would expect to see less presupposition filtering in (22a) (given that we found the effect of redundancy to be lessened in NEGFIRST conditionals – in fact there was no evidence for it, see Table 5)⁹, compared to (22b) (since the effect of redundancy were stronger in NEGSECOND conditionals). If this turns out to be the case, it will be an interesting argument for the parallelism we have been entertaining in this section.

- (22) a. If John is studying in Japan too, Mary is not studying in Tokyo.
 b. Mary is not studying in Tokyo, if John is studying in Japan too.

2014). The experiments we reference include explicit controls to rule out this potential confound (see Kalomoiros & Schwarz 2024).

⁹ Recall that a stronger effect of redundancy means lower acceptability in our tasks.

Having highlighted the attested and potential parallels between the two phenomena, we must note an important caveat. By contrast with the presupposition literature, we did not test *embeddings* of our target sentences, so it is possible that the contrast we observed between disjunction and conjunction is a discourse-level contrast. That is, perhaps $p \wedge p^+$ can be interpreted as a successive assertion, of p and then p^+ , allowing us to make sense of it pragmatically, while $p^+ \wedge p$ cannot. By contrast, no such parallel discourse-integration strategy arises in the case of disjunction. Thus, before concluding that the situations for redundancy and triviality are indeed parallel, it would be illuminating to test embeddings of our target cases in future work.¹⁰

With this in mind, let's assume for the sake of argument that there is indeed a parallel between presupposition and redundancy, and it's what accounts for the order-based contrast between conjunction and disjunction (points (ii) and (iii) in 7.1). Let's also assume that the same parallelism is what eventually explains the polarity-dependent order effect in conditionals (point (v) in 7.1): POSITIVE conditionals are redundant in both orders, but NEGATIVE conditionals are only redundant in SECOND order.¹¹ We then still need to account for the following points:

- i. POSITIVE disjunctions are more redundant than NEGATIVE disjunctions
- iv. POSITIVE disjunctions are more redundant than SECOND conjunctions.
- vi. POSITIVE disjunctions are more redundant than POSITIVE conditionals.
- vii. POSITIVE conditionals and SECOND conjunctions exhibit parallel strength of redundancy.

We now turn to how the notion of super-redundancy can help here.

¹⁰ At the same time, it seems less likely that the polarity-dependent order effect that we found for conditionals can be explained via different strategies for asserting a conditional with a positive vs a negative antecedent. Thus, if a parallel order effect is confirmed on the filtering side, this will constitute clearer evidence that filtering and redundancy are indeed parallel.

¹¹ Attributing these facts to the parallelism between filtering and redundancy is not a completely arbitrary decision. The theory in Kalomoiros (2023b) predicts exactly this constellation of facts for filtering: the presuppositions of a positive antecedent in a conditional like (22) are projected in the antecedent-first order but filtered in the antecedent-final order. Conversely, for a conditional with a negative antecedent like (21), the presuppositions of the antecedent are filtered in either order.

7.3 Super-redundancy

7.3.1 Intuitions and definition

The notion of *super-redundancy* was first developed in Kalomoiros 2023a. Kalomoiros builds on the notion of a redundant constituent from the Simplification theory of redundancy. Recall that a constituent is redundant in S if the result of deleting that constituent is contextually equivalent to S . A constituent is said to be *super-redundant* if it is not only redundant, but also irredeemably so—that is, if any attempt to strengthen the constituent still results in redundancy.

- (23) **Super-redundant constituent:** Given a sentence S , a context C , a constituent E of S is super-redundant iff for all atomic constituents E' of E , and for all D : $E[E'|E' \wedge D]$ is redundant in S' in C , where: $E[E'|E' \wedge D]$ is identical to E except that the atomic constituent E' has been replaced with $(E' \wedge D)$; and S' is a sentence identical to S but where E has been replaced by $E[E'|E' \wedge D]$.¹²

The idea is that a sentence which contains a constituent which is redundant will strike us as odd, while a sentence which contains a constituent which is redundant and also super-redundant will strike us as odder still.

7.3.2 Conjunction vs Disjunction

The notion of super-redundancy can help account for two key aspects of our results: why negative Hurford disjunctions are judged to be less odd than positive Hurford disjunctions, and why positive Hurford disjunctions are judged to be more odd than redundant conjunctions.

First, note that positive Hurford disjunctions are both redundant and super-redundant. $p^+ \vee p$ (and its reverse order) is equivalent to p , which means that p^+ is redundant here. Moreover, p^+ is super-redundant in this case: no matter what we attempt to conjoin it with, its redundancy cannot be eliminated, because $(p^+ \wedge D) \vee p$ is always equivalent to p , no matter what D is. So positive Hurford disjunctions contain a constituent which is not only redundant but irredeemably so.

By contrast, negative Hurford disjunctions are not super-redundant. $\neg p$ is redundant in $\neg p \vee \neg p^+$, because the latter is equivalent to $\neg p^+$; but it is not super-redundant, because we can select a D to conjoin with one of its atomic constituents in a way that eliminates the redundancy of the constituent. For instance,

¹² Here, we underlyingly assume a garden variety propositional language consisting of propositional constants, and complex sentences formed via negation, conjunction, disjunction and implication. As such, ‘atomic constituent’ refers to the simple propositional constants in this language.

$\neg(p \wedge \perp) \vee \neg p^+$ is not generally equivalent to $\neg p^+$ but rather to $\top$. So $\neg p$ is redundant but not irredeemably redundant in $\neg p \vee \neg p^+$.

Super-redundancy likewise lets us distinguish between the level of oddness observed in positive Hurford disjunctions vs. that seen in redundant conjunctions. We have seen that $p^+ \vee p$ is super-redundant. By contrast, $p^+ \wedge p$ is only redundant: the conjunct p is redundant, but it is not super-redundant, since, e.g., $p^+ \wedge (p \wedge \perp)$ is not generally equivalent to p^+ but rather to $\perp$.

7.3.3 Conditionals

What about the remaining effects with conditionals, i.e., why:

- vi. Pos disjunctions are more redundant than Pos conditionals.
- vii. Pos conditionals and SECOND conjunctions exhibit parallel strength of redundancy.

Let's start by briefly exploring the predictions super-redundancy makes for conditionals if we assume a simplified material implication semantics.

For now, assume that $\rightarrow$ is just the material conditional. In $(\neg p^+ \rightarrow p)$, $\neg p^+$ is redundant, since $\neg p^+ \rightarrow p$ is equivalent to p . But $\neg p^+$ is also super-redundant, since attempting to strengthen its atomic sub-constituents doesn't break the equivalence of $\neg p^+ \rightarrow p$ with p .¹³ By contrast, $p \rightarrow \neg p^+$ is not super-redundant: p is redundant, but it is not super-redundant, since, e.g., $(p \wedge \perp) \rightarrow \neg p^+$ is no longer equivalent to $\neg p^+$ but rather to $\top$.

There are three issues with this account of redundancy effects in Hurford conditionals: first it predicts that NEGATIVE and POSITIVE Hurford conditionals should contrast, with the latter being more strongly redundant than the former. As we have already seen, such a contrast holds only for ORDER = FIRST. But even in that case, it's not clear that that the contrast was due to the fact that NEGFIRST conditionals were redundant, and POSFIRST conditionals were redundant to an even greater extent – in fact there is no evidence for the presence of a redundancy violation in NEGFIRST conditionals in the first place.

The other two issues relate to how connectives should differ with respect to their redundancy violations. Indeed, Super-redundancy predicts that NEGATIVE and

¹³ Note that the requirement that atomic sub-constituents of $\neg p^+$ be the ones subject to strengthening is crucial. If we allowed a redundant constituent E to escape super-redundancy simply because any strengthening of any of its sub-constituents (regardless of their granularity) made E non-redundant, then $\neg p^+$ would no longer count as super-redundant in $\neg p^+ \rightarrow p$ since e.g. $(\neg p^+ \wedge \perp) \rightarrow p$ is equivalent to $\top$, not (generally) to p .

POSITIVE conditionals should pattern like their corresponding disjunctions. While we saw that this was the case for NEGATIVE conditionals (modulo the discussion in the previous paragraph), we found a contrast between POSITIVE conditionals and POSITIVE disjunctions: the latter showed a stronger effect of redundancy than the former. Relatedly, POSITIVE conditionals are predicted to be more redundant than SECOND conjunctions: the former are super-redundant while the latter only redundant. But we found no significant contrast in this respect.

Thus, to maintain an account of conditionals along the lines predicted by super-redundancy, one would have to come up with an explanation as to why POSITIVE conditionals are not as redundant as POSITIVE disjunctions. This explanation should also provide an answer to why POSITIVE conditionals are statistically not distinguishable from SECOND conjunctions in terms of redundancy. We do not currently have such an answer, so for the time being we leave this as a puzzle.

Having said that, we also acknowledge that the above account of Hurford conditionals is only a promissory note, since the natural language conditional cannot realistically be treated as the material conditional. Moreover, super-redundancy must be eventually integrated into a framework that can handle both presupposition and Hurford phenomena. As such, future research should explore whether making the requisite moves of adopting a more plausible conditional semantics and better integrating redundancy and presupposition might hold solutions to the remaining puzzles we have outlined here.

8 Conclusion

We have presented three experiments that advance our understanding of redundancy in multiple ways. On one hand, our results validate a key empirical observation motivating much of this literature: Hurford disjunctions, conditionals, and redundant conjunctions are generally judged to be odd. On the other hand, they unveil a range of additional effects—related to order, polarity, and the choice of connective—that are not straightforwardly predicted by standard Simplification or Local-Context approaches to redundancy. We have discussed two (currently partial) accounts of these phenomena: one linking order effects to analogous patterns in presupposition filtering, and another relating differences in the strength of redundancy to the notion of super-redundancy. While these proposals account for a substantial portion of the effects we observed, several theoretical and empirical questions remain, particularly concerning how (super-)redundancy operates in conditionals. We suggested that further progress may come from investigating how constraints on the redundancy and triviality of at-issue content can be integrated with constraints governing presupposition.

References

- George, Benjamin. 2007. Predicting presupposition projection: some alternatives in the strong Kleene tradition. Manuscript, UCLA. <http://semanticsarchive.net/Archive/DY0YTgxN/>.
- George, Benjamin R. 2008. A new predictive theory of presupposition projection. In *Semantics and Linguistic Theory*, vol. 18, 358–375.
- Hirsch, Aaron & Martin Hackl. 2014. Incremental presupposition evaluation in disjunction. In *Proceedings of the 44th annual meeting of the North East Linguistic Society*, vol. 1, 177–190.
- Hurford, James. 1974. Exclusive or inclusive disjunction. *Foundations of Language* 11. 409–411.
- Kalomoiros, Alexandros. 2023a. An approach to hurford conditionals. In *Proceedings of salt 33*, 724–743. Heinrich Heine Universität, Düsseldorf: Linguistic Society of America. <http://elanguage.net/journals/index.php/salt>.
- Kalomoiros, Alexandros. 2023b. *Presupposition and its (A-)symmetries*: University of Pennsylvania PhD dissertation.
- Kalomoiros, Alexandros, Matthew Mandelkern, Jacopo Romoli & Florian Schwarz. 2024. Presuppositions project asymmetrically, unless they don't. In *Proceedings of the Amsterdam Colloquium 2024*, 196–202.
- Kalomoiros, Alexandros & Florian Schwarz. 2024. Presupposition projection from 'and' vs 'or': Experimental data and theoretical implications. *Journal of Semantics*.
- Karttunen, Lauri. 1974. Presupposition and linguistic context. *Theoretical Linguistics* 1(1-3). 181–194.
- Katzir, Roni & Raj Singh. 2013. A note on presupposition accommodation. *Semantics and Pragmatics* 6. 5–1.
- Mandelkern, Matthew & Jacopo Romoli. 2018. Hurford Conditionals. *Journal of Semantics* 35(2). 357–367. <https://doi.org/10.1093/jos/ffx022>.
- Mandelkern, Matthew, Jérémy Zehr, Jacopo Romoli & Florian Schwarz. 2020. We've discovered that projection across conjunction is asymmetric (and it is!). *Linguistics and Philosophy* 43(5). 473–514.
- Marty, Paul & Jacopo Romoli. 2022. Varieties of hurford disjunctions. *Semantics and Pragmatics* 15. 3–1.
- Mayr, Clemens & Jacopo Romoli. 2016. A puzzle for theories of redundancy: Exhaustification, incrementality, and the notion of local context. *Semantics and Pragmatics* 9. 1–48.
- Schlenker, Philippe. 2008. Be articulate: A pragmatic theory of presupposition projection. *Theoretical Linguistics* 157–212.
- Schlenker, Philippe. 2009. Local contexts. *Semantics and Pragmatics* 2. 1–78.

- Stalnaker, Robert. 1974. Pragmatic presuppositions. In Milton K. Munitz & Peter K. Unger (eds.), *Semantics and Philosophy*, 197–213. New York University Press.
- Stalnaker, Robert C. 1978. Assertion. In Peter Cole (ed.), *Syntax and Semantics 9*, 315–332. Brill.